The Conversion of Aluminum Foil to Potassium Aluminum Sulfate Dodecahydrate Using Acid-Base Neutralization Reactions and Physical Separation of Solutions

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CHM 113 L1: Elements and Compounds Laboratory

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November 6, 2023

Abstract

Aluminum foil was converted into potassium aluminum sulfate dodecahydrate. The aluminum was oxidized to aluminum (III) using potassium hydroxide, which was the source of potassium ions. The resulting aluminate ion was neutralized with sulfuric acid. The new solution was filtered to form dry alum (79.96 percent yield), and tested for component ions: potassium (negative), sulfate (positive), and aluminum (positive). The melting point range was determined (83.1°C - 88.7°C) and compared to commercially produced alum (84.2°C - 92.4°C).

Introduction

Alum is a substance that usually consists of aluminum sulfate, water of hydration, and a sulfate of another element.[1] The most common type of alum is known as potassium aluminum sulfate dodecahydrate, or potassium alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$)[2], which is a white powdery substance that dissolves in water.[3]

There are a variety of uses of potassium alum, mainly found in the agricultural world. Potassium alum serves as a catalyst for the extraction of essential oils in pine which is used in pharmaceuticals, flavors, and fragrances.[4] It also strengthens the stability of starch molecules in potatoes[5] and is used in water treatment to preserve organic materials and immobilize metals.[6] Potassium alum can also be used industrially to size papers and reinforce dyes in clothing.[7]

In an industrial setting, bauxite is treated with sulfuric acid in order to make aluminum sulfate, which is then combined with potassium sulfate to make potassium alum.[8] In this experiment, a sample of aluminum foil is converted into potassium alum through a series of reactions that involve the neutralization of aluminum hydroxide and sulfuric acid.

Aluminum is oxidized to aluminum (III) using potassium hydroxide as in Equation 1. This produced potassium and aluminate ions in solution with a release of hydrogen gas.

$$2\ KOH\ (aq) + 2\ Al\ (s) + 6\ H2O\ (\ell) \rightarrow 2\ KAl(OH)_4\ (aq) + 3\ H_2\ (g) \qquad \textbf{(1)}$$

Sulfuric acid is added to the new solution to neutralize a hydroxide from the aluminate ion, shown in net ionic Equation 2, resulting in the formation of crystals containing K^+, Al^{3+}, and SO_4^{2-} after the reaction in net ionic Equation 3. These crystals, formed during purification by recrystallization, contain water molecules of hydration, thus forming potassium alum.

$$2\ Al(OH)_{4-}\ (aq) + H_2SO_4\ (aq) \rightarrow 2\ Al(OH)_3\ (s) + 2\ H_2O\ (\ell) + SO_4^{2-}\ (aq) \qquad \textbf{(2)}$$

$$2\ Al(OH)_3\ (s) + 3\ H_2SO_4\ (aq) \rightarrow 2\ Al^{3+}\ (aq) + 6\ H_2O\ (\ell) + 3\ SO_4^{2-}\ (aq) \qquad \textbf{(3)}$$

The potassium alum was tested for the presence of potassium, sulfate, and aluminum ions. The flame test can be used to determine the presence of potassium ions, as potassium displays a purple flame when burned. Barium is used to test for sulfate ions. Barium sulfate is an insoluble compound. Therefore, the addition of barium chloride results in a white precipitate in the presence of sulfate. Aluminon solution can be used to test for the presence of aluminum ions, since aluminon reacts with aluminum (III) to form a pink solution. Finally, the melting point of the alum can be determined and compared to a commercial sample.

Procedure

The experiment was conducted using the procedure originally documented in the Elements and Compounds Laboratory Manual by Dr. Megan Youmans of the Department of Chemistry at Wilkes University.[9] A square-shaped sample of aluminum was weighed to a mass of 0.4991 grams and placed in a clean 100 mL beaker. A portion of 24.8 mL of 4.0 *M* KOH was slowly added to the aluminum. The reaction was stirred until the solution turned an opaque gray with effervescence. The reaction was heated and stirred until all gas evolution ceased and the aluminum was completely reacted, resulting in a darker gray solution. The reaction was gravity filtered into a 50 mL beaker and 15.1 mL of 9 *M* H_2SO_4 were slowly added to the filtrate, which initially formed a thick white precipitate and an opaque white solution upon completion. The reaction was placed in an ice bath for 15 minutes. The crystals were collected in a vacuum and washed with cold ethanol. The beaker was rinsed three times with three mL ethanol and the vacuum was allowed to continue until the Buchner funnel contained a coarse white powder. A pre-massed paper cup was used to contain the filtered substance and the product was allowed to dry open to air for one week.

The mass of synthesized alum was determined with difference and small portions were used in four different qualitative tests. A flame test was performed on a small sample of the alum alongside a positive control (KCl crystals) and negative control (NaCl crystals), resulting in a bright orange flame for the alum and negative control and a faint purple for the positive control. An aluminon test was performed using some of the alum, a positive control ($AlCl_3$), and an negative control (NaCl). Three test tubes were prepared with distilled water, four drops of 80 mM aluminon solution, and a pinch of one test substance to total 10 mL in volume. The tubes were mixed and allowed to sit for twenty minutes before examination of color, resulting in a transparent pink solution for the alum and positive control and no color for the negative control. A sulfate test was done in three smaller test tubes, each with a pinch of one of three substances: alum, a positive control (Na_2SO_4), and a negative control (NaCl). A small volume of distilled water was added to each tube to allow substances to dissolve before adding one drop of 0.5 *M* $BaCl_2$, resulting in the formation of a white precipitate in the alum and positive control and no changes in the negative control.

Results

The mass of the synthesized alum was determined by finding the difference in mass of the full paper cup and the contents. The theoretical yield of alum was determined using a stoichiometric conversion. Table I contains numerical data for the mass, theoretical yield, and the percent yield of alum.

Table I: Determination of Mass of Alum Produced

Mass of aluminum foil (g)	0.4991
Mass of filter paper (g)	0.2901
Mass of paper cup (g)	2.6468
Mass of paper cup with filter paper and alum crystals (g)	9.9530
Mass of alum (g)	7.0161
Theoretical yield of alum (g)	8.775
Percent yield (%)	79.96

The produced alum was examined through the qualitative tests listed in Table II. The result of each test was compared to the indicated positive and negative controls.

Table II: Qualitative Testing of Synthesized Alum

Test Performed	Control (+)	Control (-)	Result (Alum)
Potassium Flame Test (K^+)	KCl (purple)	NaCl (orange)	Negative (orange)
Sulfate Test (precipitation of $BaSO_4$)	Na_2SO_4 (white ppt)	NaCl (no ppt)	Positive (white ppt)
Colorimetric Aluminon Test (Al^{3+})	$AlCl_3$ (pink)	NaCl (colorless)	Positive (pink)

The initial temperature at which the experimental alum sample began to melt and the final temperature at which the sample was completely melted were both recorded to determine the melting point range. A commercial standard was also measured to account for errors in the thermometer or judgment of Mel-Temp readings. The range for the synthesized alum (83.1°C - 88.7°C) and the commercial standard (84.2°C - 92.4°C) were compared to the accepted melting point of alum (92.5°C).[10] The final temperature at which both were completely melted were used to calculate percent error, where the synthesized alum was erroneous with 4.11 percent error, and the commercial standard was more accurate with 0.11 percent error.

Discussion

The formation of potassium alum from a sample of aluminum foil requires the addition of potassium and sulfate ions. Equation 1 details how potassium hydroxide was reacted with the aluminum sample in a solution, yielding hydrogen gas and aluminate ions. This reaction is classified as a redox reaction. Aluminum is oxidized to aluminum (III) and water is reduced.

Using another base for this reaction would not be sufficient, as the potassium ions are required for the synthesis of alum. The resultant potassium aluminate was neutralized by sulfuric acid as in Equation 2. This caused the formation of a white precipitate, aluminum hydroxide. The use of sulfuric acid is necessary to obtain sulfate ions used in the formation of alum. The final reaction in Equation 3 neutralized aluminum hydroxide to form aluminum ions and water. The resulting solution was recrystallized to remove impurities, and then dried in open air. The product was isolated with 79.96 percent yield. The mass of alum may have been lost during the neutralization reactions, where some ions were filtered with excess water. Part of the mass was likely lost during the transfer of solutions from one container to another. Another potential source of error is the mass balance, resulting in inaccuracy of mass readings for components recorded in Table I. Finally, the mass of alum was decreased during the recrystallization process. The solubility of alum is 1 g/7.5 mL at room temperature[10], so the solution may not have been saturated and all of the alum did not isolate completely.

The synthesized alum was tested for ions. In the flame test, the alum burned an orange color like that of the negative control, NaCl. This indicates that the alum sample did not contain potassium ions. However, it is possible that both sodium and potassium were present. The orange flame caused by sodium may have subjugated the faint purple flame of the potassium. The lack of potassium could be explained by the crystallization process used to purify alum, where excess ions were discarded in filtrate. The alum formed a white precipitate during the sulfate test, which indicates the presence of sulfate ions. The alum sample also tested positive for aluminum ions in the aluminon solution test, as it formed the same pink color as the positive control, $AlCl_3$.

The water of hydration was never tested in this procedure. It can be quantified with a distillation filter setup, where the hydration molecules can be collected in a separate chamber after being isolated from alum. Finally, the melting point range was examined for the synthesized alum inside a Mel-Temp device. The final temperature at which the alum completely melted was used to determine percent error, where the accepted melting point was 92.5°C[10]. The percent error calculated was 4.11 percent. This deviation is likely due to the inaccurate yield of alum and the lack or surplus of potassium, sulfate, or aluminum ions, which were never determined by quantity. Another source of error could be the calibration of the thermometer or power on the Mel-Temp device. The commercial standard used to determine error in the device presented 0.11 percent error. Test tubes used in the sulfate and aluminum tests may have contained residual substances from other procedures. Finally, human error contributed to the reason why the exact melting temperature range may not have been accurate. This experiment can be improved by using a sample of commercially produced pure aluminum. The filtrate from the vacuum filtration could be used in another precipitation reaction in order to prevent the loss of filtered ions. In general, the equipment and filtration methods could be improved to be more rigorous and safeguard against further systematic errors.

Conclusion

Alum was synthesized from aluminum foil in 79.96 percent yield. The presence of component ions were confirmed with qualitative tests. Potassium ions were tested using a flame test, aluminum ions were tested in aluminon solution, and sulfate ions were tested with barium chloride. The melting point of alum was 83.1°C - 88.7°C, which was lower compared to a commercial alum standard at 84.2°C - 92.4°C.

References

1. Encyclopedia Britannica. *Alum*. Credoreference.com. https://search.credoreference.com/articles/Qm9va0FydGljbGU6NDM5NzA3?q=alum (accessed 2023-11-06).
2. ChEBI. *Potassium Aluminium Sulfate Dodecahydrate (CHEBI:86465)*. www.ebi.ac.uk. https://www.ebi.ac.uk/chebi/searchId.do?chebiId=CHEBI:86465#:~:text=Potassium%20alum%2C%20potash%20alum%2C%20or (accessed 2023-11-06).
3. The Hutchinson Unabridged Encyclopedia with Atlas and Weather Guide. *Alum*. Credoreference.com. https://search.credoreference.com/articles/Qm9va0FydGljbGU6NDgxNzQ1?q=uses%20of%20alum (accessed 2023-11-06).
4. Wijayati, N.; Lestari, L. R.; Wulandari, L. A.; Mahatmanti, F. W.; Rakainsa, S. K.; Cahyono, E.; Wahab, R. A. Potassium Alum [KAl(SO4)2·12H2O] Solid Catalyst for Effective and Selective Methoxylation Production of Alpha-Pinene Ether Products. *Heliyon* 2021, *7* (1), e06058. https://doi.org/10.1016/j.heliyon.2021.e06058.
5. Li, Z.; Zhang, Y.; Ai, Z.; Fan, H.; Wang, N.; Suo, B. Effect of Potassium Alum Addition on the Quality of Potato Starch Noodles. *Journal of Food Science and Technology* **2019**, *56* (6), 2932–2939. https://doi.org/10.1007/s13197-019-03760-3.
6. Haynes, R. J.; Zhou, Y. F. *Use of alum water treatment sludge to stabilize C - ProQuest*. www.proquest.com. https://www.proquest.com/docview/1788586608/fulltextPDF/E065941DBAA44E88PQ/1?accountid=62703 (accessed 2023-11-06).

7. The Editors of Encyclopedia Britannica. Alum | Chemical Compound. *Encyclopædia Britannica*; 2023.
8. Affinity Chemical. *Alum Manufacturing Techniques & the Affinity Process*. Affinity Chemical. https://www.affinitychemical.com/alum-manufacturing-techniques-the-affinity-process/#:~:text=From%20Bauxite&text=Adding%203%20molecules%20of%20sulfuric (accessed 2023-11-06).
9. Youmans, M. *Conversion of an Aluminum Can to (Potassium) Alum;* Wilkes University Department of Chemistry and Biochemistry: Wilkes-Barre, 2023 (accessed 2023-10-30).
10. PubChem. *Aluminum potassium sulfate*. Pubchem.ncbi.nlm.nih.gov. https://pubchem.ncbi.nlm.nih.gov/compound/Aluminum-potassium-sulfate#section=Physical-Description (accessed 2023-11-06).

Appendix

The theoretical yield recorded in Table I was calculated using the following formula, where the value T is the theoretical yield and the value m_a is the mass of the aluminum foil:

$$T = m_a\left(\frac{mol\ Al}{26.982\ g\ Al}\right)\left(\frac{mol\ Alum}{mol\ Al}\right)\left(\frac{474.37\ g\ Alum}{mol\ Alum}\right)$$

$$T = (0.4991\ g\ Al)\left(\frac{mol\ Al}{26.982\ g\ Al}\right)\left(\frac{mol\ Alum}{mol\ Al}\right)\left(\frac{474.37\ g\ Alum}{mol\ Alum}\right) = 8.775\ g\ Alum$$

The percent yield (P) recorded in Table I was found with the percent yield formula:

$$P = \frac{actual\ yield}{theoretical\ yield} \times 100\%$$

$$P = \frac{7.0161\ g\ Alum}{8.775\ g\ Alum} \times 100\% = 79.96\%$$

The percent error (P_E) recorded in Table III was found with the percent error formula. The accepted melting point of potassium alum is 92.5°C.[12]

$$P = \frac{|experimental - actual|}{actual} \times 100\%$$

$$P = \frac{|88.7 - 92.5|}{92.5} \times 100\% = 4.11\% \text{ (synthesized alum)}$$

$$P = \frac{|92.4 - 92.5|}{92.5} \times 100\% = 0.11\% \text{ (commercial alum)}$$